Department of Mathematics, IIT Madras
MA2031-Linear Algebra for Engineers
Assignment-I

1. In each of the following a nonempty set V is given and some operations are defined. Check whether V is a vector space with these operations. Whenever V is a vector space, write explicitly the zero vector and the additive inverse of any vector $v \in V$.
   (a) $V = \{(a,0) : a \in \mathbb{R}\}$ with $+$ and $\cdot$ as in $\mathbb{R}^2$.
   (b) $V = \{(a,b) \in \mathbb{R}^2 : 2a+3b = 0\}$ with $+$ and $\cdot$ as in $\mathbb{R}^2$.
   (c) $V = \{(a,b) \in \mathbb{R}^2 : a+b = 1\}$ with $+$ and $\cdot$ as in $\mathbb{R}^2$.
   (d) $V = \mathbb{R}^2$ with $+$ as in $\mathbb{R}^2$, and $\cdot$ defined by $0 \cdot (a,b) = (0,0)$, and for $\alpha \neq 0$, $\alpha \in \mathbb{R}$, $\alpha \cdot (a,b) = (a/\alpha, \alpha b)$.
   (e) $V = \mathbb{R}^2$ with $+$ as in $\mathbb{R}^2$, and $\cdot$ defined by $\alpha \cdot (a,b) = (a,0)$ for $\alpha \in \mathbb{R}$.
   (f) $V = \mathbb{C}^2$ with $+$ and $\cdot$ defined by $(a,b)+(c,d) = (a+2c, b+3d)$, and $\alpha \cdot (a,b) = (\alpha a, \alpha b)$ for (a,b), $(c,d) \in V$ and $\alpha \in \mathbb{C}$.
   (g) $V = \mathbb{R}_+$, the set of all positive real numbers, with addition $\oplus$ and scalar multiplication $\odot$ defined by $x \oplus y = xy$, and $\alpha \odot x = x^\alpha$ for $x, y \in V$ and $\alpha \in \mathbb{R}$.
   (h) $V = \mathbb{R}_+ \cup \{0\}$ with addition $\oplus$ and scalar multiplication $\odot$ defined by $x \oplus y = xy$, and $\alpha \odot x = |\alpha| x$ for $x, y \in V$ and $\alpha \in \mathbb{R}$.
   (i) $V = \mathbb{R} \times \mathbb{R}_+$, with addition $\oplus$ as $(a,b) \oplus (c,d) = (a+c, bd)$ and scalar multiplication $\odot$ defined by $\alpha \odot (a,b) = (\alpha a, b^\alpha)$ for $\alpha \in \mathbb{R}$.
   (j) $V = \mathbb{Q}$, the set of all rational numbers, with $+$ and $\cdot$ as in $\mathbb{R}$.
   (k) $(\mathbb{R} \setminus \mathbb{Q}) \cup \{0,1\}$ with $+$ and $\cdot$ as in $\mathbb{R}$.
   (l) $V = \mathbb{R}$ with addition $\oplus$ as $a \oplus b = a+b-2$ and the scalar multiplication defined by $\alpha \odot b = \alpha b + 2(\alpha - 1)$ for $\alpha \in \mathbb{R}$.
   (m) $V = \mathbb{R}^n$ with addition $\oplus$ as $u \oplus v = u+v-w$ and scalar multiplication $\odot$ as $\alpha \odot u = \alpha(u-w)+w$, where w is a fixed given vector in V, and $\alpha \in \mathbb{R}$, $u, v \in V$.
   (n) $V = \{(a,b) \in \mathbb{R}^2 : a+b = 1\}$ with addition $\oplus$ and scalar multiplication $\odot$ as $(a,b) \oplus (c,d) = (a+c-1, b+d)$, and $\alpha \odot (a,b) = (\alpha a - \alpha + 1, \alpha b)$. Contrast it with (c).

2. Is the set of all purely imaginary numbers a real vector space with usual addition and scalar multiplication?

3. Is the set of all $n \times n$ hermitian matrices a real vector space?

4. Is the set of all $n \times n$ hermitian matrices a complex vector space?

5. In each of the following, a vector space V and a subset U of V are given. Check whether U is a subspace of V.
   (a) $V = \mathbb{R}^2$, $U = \{(a,b) \in V : b = 2a - \alpha\}$ for some $\alpha \neq 0$.
   (b) $V = \mathbb{R}^3$, $U = \{(a,b,c) \in V : 2a - b - c = 0\}$.
   (c) $V = \mathbb{F}_3[t]$, $U = \{a+bt+ct^2+dt^3 \in V : \ a = 0\}$.
   (d) $V = \mathbb{C}_3[t]$, $U = \{a+bt+ct^2+dt^3 \in V : a+b+c+d = 0\}$.
   (e) $V = \mathbb{C}_3[t]$, $U = \{a+bt+ct^2+dt^3 \in V : a,b,c,d \text{ integers}\}$.
   (f) $V = C[-1,1]$, $U = \{f \in V : f \text{ is an even function}\}$.
   (g) $V = C[0,1]$, $U = \{f \in V : f(t) \geq 0 \text{ for all } t \in [0,1]\}$.
   (h) $V = \mathbb{R}^{n \times n}$, $U = \{A \in V : A^{t} = A\}$.
   (i) $V = \mathbb{C}^{n \times n}$, $U = \{A \in V : A^* = A\}$.

6. Describe all subspaces of $\mathbb{R}^2$, and of $\mathbb{R}^3$.

7. Let $U_b = \{(a_1, a_2, \ldots, a_n) : a_1, a_2, \ldots, a_n \in \mathbb{R},\ a_1 + 2a_2 + \cdots + na_n = b\}$. For which real numbers b, is U_b a subspace of $\mathbb{R}^n$?
8. Let U be a subspace of V and let V be a subspace of W. Is U a subspace of W?
9. Is the set of all skew-symmetric $n \times n$ matrices a subspace of $\mathbb{R}^{n\times n}$?
10. Determine whether the following are real vector spaces:
    (a) $\mathbb{R}^\infty$:= the set of all sequences of real numbers.
    (b) ℓ^∞ := the set of all bounded sequences of real numbers.
    (c) ℓ^1 := the set of all absolutely convergent sequences of real numbers.
11. Let S be a nonempty set and let $s \in S$. Let V be the set of all functions $f : S \to \mathbb{R}$ with $f(s) = 0$. Is V a vector space over $\mathbb{R}$ with the usual addition and scalar multiplication of functions?
12. Show that the set $B(S)$ of all bounded functions from a nonempty set S to $\mathbb{R}$ is a real vector space.
13. Do the polynomials $t^3 - 2t^2 + 1$, $4t^2 - t + 3$, and $3t - 2$ span $\mathbb{F}_3[t]$?
14. What is span $\{t^n : n = 0, 2, 4, 6, \ldots\}$ in $\mathbb{R}[t]$?
15. In $\mathbb{F}^3$, let $e_1 = (1,0,0)$, $e_2 = (0,1,0)$ and $e_3 = (0,0,1)$. span $\{e_1 + e_2, e_2 + e_3, e_3 + e_1\}$ =?
16. Let $u, v_1, v_2, \ldots, v_n$ be $n+1$ distinct vectors in a vector space V. Take $S_1 = \{v_1, v_2, \ldots, v_n\}$ and $S_2 = \{u, v_1, v_2, \ldots, v_n\}$. Prove that $u \in \operatorname{span}(S_1)$ iff $\operatorname{span}(S_1) = \operatorname{span}(S_2)$.
17. Let S be a subset of a vector space V. Show that S is a subspace iff $S = \operatorname{span}(S)$.
18. Let $u_1(t) = 1$, and for $n = 2, 3, \ldots$, let $u_n(t) = 1 + t + \ldots + t^{n-1}$. Show that $\{u_1, \ldots, u_n\}$ spans $\mathbb{F}_{n-1}[t]$. Is it true that $\{u_1, u_2, \ldots\}$ spans $\mathbb{F}[t]$?
19. Let S be a subset of a vector space V. Prove that span (S) is the intersection of all subspaces that contain S.
20. We know that $e^t = 1 + t + \frac{1}{2!}t^2 + \cdots$ for each $t \in \mathbb{R}$. Does it imply that $e^t \in \operatorname{span}\{1, t, t^2, \ldots\}$?
21. Show that every vector space has at least two spanning sets.
22. Let V be the real vector space of all functions from $\{1, 2\}$ to $\mathbb{R}$. Construct a spanning set of V with two elements.
23. Find a finite spanning set for the space of all real symmetric matrices of order n.
24. Let A and B be subsets of a vector space V. Prove (a)-(c) and give counter examples for (d)-(f):
    (a) $\operatorname{span}(\operatorname{span}(A)) = \operatorname{span}(A)$.
    (b) If $A \subseteq B$, then $\operatorname{span}(A) \subseteq \operatorname{span}(B)$.
    (c) $\operatorname{span}(A \cap B) \subseteq \operatorname{span}(A) \cap \operatorname{span}(B)$.
    (d) $\operatorname{span}(A) \cap \operatorname{span}(B) \subseteq \operatorname{span}(A \cap B)$.
    (e) $\operatorname{span}(A) \setminus \operatorname{span}(B) \subseteq \operatorname{span}(A \setminus B)$.
    (f) $\operatorname{span}(A \setminus B) \subseteq (\operatorname{span}(A) \setminus \operatorname{span}(B)) \cup \{0\}$.
25. Give suitable real vector spaces U, V, W so that $U + V = U + W$ but $V \neq W$.
26. Let U and W be subspaces of a vector space such that $U \cap W = \{0\}$. Prove that if $x \in U + W$, then there exist unique $u \in U, w \in W$ such that $x = u + w$.
27. Let U, V, and W be subspaces of a vector space X.
    (a) Prove that $(U \cap V) + (U \cap W) \subseteq U \cap (V + W)$.
    (b) Give suitable U, V, W, X so that $U \cap (V + W) \nsubseteq (U \cap V) + (U \cap W)$.
    (c) Prove that $U + (V \cap W) \subseteq (U + V) \cap (U + W)$.
    (d) Give suitable U, V, W, X so that $(U + V) \cap (U + W) \nsubseteq U + (V \cap W)$.